

Chapter 8 Notes

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1. **Theorem 8.3.** Let (a_{ij}) be a double sequence,

$$\sum_{j=1}^{\infty} |a_{ij}| = b_i,$$

and $\sum b_i$ converge. Then

$$\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} a_{ij} = \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} a_{ij}.$$

2. **Theorem 8.5.** Let $\sum a_n x^n$ and $\sum b_n x^n$ converge on $S = (-R, R)$. Define

$$E = \left\{ x \in S \mid \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} b_n x^n \right\}.$$

Then if $E' \subseteq S$, then $a_n = b_n$.

3. **Definition 8.9.** Let $(a_N), (b_N) \in \mathbb{C}$. Then a *trigonometric polynomial* f is a finite sum of the form

$$\forall x \in \mathbb{R} \quad f(x) = a_0 + \sum_{n=1}^N a_n \cos(nx) + b_n \sin(nx)$$

and write $f \in \mathcal{T}$. ($\mathcal{T}$ is the set of all trigonometric polynomials.) However, the form

$$\forall x \in \mathbb{R} \quad f(x) = \sum_{n=-N}^N c_n e^{inx}$$

is often more convenient; furthermore,

$$c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx. \quad (1)$$

$f \in \mathcal{T}$ is real if and only if $c_{-n} = \overline{c_n}$. The *trigonometric series* is a series of the form

$$\forall x \in \mathbb{R} \quad f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx}. \quad (2)$$

Let (c_n) be as in (1). Then (c_n) is said to be the *Fourier coefficients* of f and (2) the *Fourier series* of f provided $f \in \mathcal{R}$ on $[-\pi, \pi]$. (Note: The book does not use the notation $\mathcal{T}$ to denote the set of all trigonometric polynomials.)

4. **Definition 8.10.** Let $(\phi_n) : [a, b] \rightarrow \mathbb{C}$ such that

$$\int_a^b \phi_n(x) \overline{\phi_m(x)} dx = 0 \quad (n \neq m).$$

Then (ϕ_n) is said to be an *orthogonal system of functions* on $[a, b]$; furthermore, (ϕ_n) is said to be *orthonormal* provided

$$\forall n \in \mathbb{N} \quad \int_a^b |\phi_n(x)|^2 dx = 1.$$

In addition, c_n is said to be the *n th Fourier coefficient of f relative to (ϕ_n)* provided (ϕ_n) is orthonormal on $[a, b]$ and

$$c_n = \int_a^b f(t) \overline{\phi_n(t)} dt;$$

furthermore,

$$f(x) \sim \sum_{n=1}^{\infty} c_n \phi_n(x)$$

is said to be the *Fourier series* of f relative to (ϕ_n) .

5. **Theorem 8.12.** Let (ϕ_n) be orthonormal on $[a, b]$ and

$$f(x) \sim \sum_{n=1}^{\infty} c_n \phi_n(x).$$

Then

$$\sum_{n=1}^{\infty} |c_n|^2 \leq \int_a^b |f(x)|^2 dx.$$

6. **Definition 8.13** (Dirichlet Kernel).

$$D_N(x) = \sum_{n=-N}^N e^{inx} = \frac{\sin((N + \frac{1}{2})x)}{\sin(\frac{x}{2})}$$

7. **Theorem 8.15.** Let f be continuous and have period 2π . Then

$$(\forall \varepsilon > 0)(\exists P \in \mathcal{T})(\forall x \in \mathbb{R}) \quad |P(x) - f(x)| < \varepsilon.$$

8. **Theorem 8.16.** Let $f, g \in \mathcal{R}$ with period 2π ,

$$f(x) \sim \sum_{n=-\infty}^{\infty} a_n e^{inx},$$

and

$$g(x) \sim \sum_{n=-\infty}^{\infty} b_n e^{inx}.$$

Then

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) \overline{g(x)} dx = \sum_{n=-\infty}^{\infty} a_n \bar{b}_n$$

and

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} |f(x)|^2 dx = \sum_{n=-\infty}^{\infty} |a_n|^2.$$

9. **Definition 8.17** (Gamma Function).

$$\forall x \in \mathbb{R}_{>0} \quad \Gamma(x) = \int_0^{\infty} t^{x-1} e^{-t} dt.$$

10. **Theorem 8.18.**

$$(a) \quad \forall x \in \mathbb{R}_{>0} \quad \Gamma(x+1) = x\Gamma(x).$$

$$(b) \quad \forall x \in \mathbb{N} \quad \Gamma(x+1) = x!.$$

11. **Theorem 8.20** (Beta Function).

$$\forall x, y \in \mathbb{R}_{>0} \quad B(x, y) = \int_0^1 t^{x-1} (1-t)^{y-1} dt = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}.$$